

Research article

Feasibility study of plasma pyrolysis on dairy waste

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ABSTRACT

Considering the ever-increasing necessity of the disposal of industrial and household waste and using mechanisms through which the utmost potential of the waste is directed to the efficient recycling of materials or forms of energy, the current study aims to investigate plasma pyrolysis as an eco-friendly and appropriate approach that leads to the production of value-added products. The novelty of this study lies in its development of a method for spoiled milk disposal since no efficient solution has been provided yet in this respect. The other benefit of implementing the proposed disposal method is the production of high-quality syngas. The maximum produced hydrogen percentage and syngas rate during the process under study in this research were obtained as 60.396 % and 2.166, respectively. A plasma method is a rare approach to yielding products with such values. The obtained results revealed that the transferred thermal plasma could efficiently dispose of the spoiled milk waste and be used as a novel approach to the green synthesis of hydrogen through plasma pyrolysis.

1. Introduction

Plasma technology exhibits a broad spectrum of applications across diverse fields. It arises as the fourth state of matter when gas undergoes energization to the extent that atomic electrons disassociate from specific atomic nuclei [1]. Some applications of plasma technology in different fields are electronics and semiconductors [2], medicine [3], energy [4], aerospace [5], material science [6,7], agriculture [8], textiles and fabrics [9], biotechnology [10], food industry [11], automotive industry [12], cosmetics and personal care [13], research and development [14,15], telecommunications [2], art conservation [16], space exploration [17], environment and pollution control [18–20] etc. The complexities of waste disposal challenges at a global level entail substantial environmental, social, and economic implications [21]. Inadequate disposal methods of waste contribute to the deterioration of the environment and contribute to climate change. The escalating global populace and advancing industrialization perpetuate a continual rise in the worldwide volume of waste. Numerous geographic areas suffer from insufficient recycling facilities and infrastructure, giving rise to inappropriate disposal methods like open dumping or incineration of waste. These practices can adversely impact both human health and the environment. In various locations, landfills are approaching their maximum capacity. Inadequately administered landfills have the potential to induce soil and water contamination, along with emissions of greenhouse gases [22]. Improper waste management plays a role in the exhaustion of valuable resources. The implementation of recycling and appropriate disposal methodologies serves to

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reclaim materials and alleviate the burden on natural resources. An increasing focus is placed on transitioning towards a circular economy [23], characterized by product designs that prioritize reuse, recycling, and minimal environmental footprint. This strategy is directed at mitigating the overall generation of waste. Eco-friendly waste disposal methods [24] play a role in mitigating the detrimental effects of waste on ecosystems, air quality, soil integrity, and water systems. Consequently, these practices contribute to biodiversity conservation and the environment's holistic well-being. Environmentally sustainable methodologies mitigate atmospheric pollution resulting from incineration, preclude soil and water contamination from landfills, and attenuate the discharge of harmful substances into the environment. Sustainable waste disposal contributes to the protection of public health [25] by reducing exposure to hazardous substances released due to inadequate waste management practices.

Traditional waste management methods, including landfilling, incineration, and composting, have been widely employed to dispose of municipal and industrial waste [26]. Nonetheless, these approaches present considerable environmental, health, and economic disadvantages. In contrast, Plasma Pyrolysis Technology (PPT) [27,28], a cutting-edge waste treatment method, provides numerous advantages over these conventional practices [19,20]. Landfills generate substantial quantities of methane, a potent greenhouse gas contributing to climate change. Leachate, a liquid byproduct of landfills, can contaminate soil and groundwater, posing serious environmental and health hazards.

Additionally, landfills require extensive land areas, which is problematic in densely populated regions. As waste production escalates, the demand for landfill space increases, exacerbating land scarcity issues. Furthermore, landfills can emit foul odors and detract from the aesthetics of the surrounding areas, leading to decreased property values and diminished quality of life for nearby residents [29]. Incineration emits harmful pollutants, such as dioxins, furans, particulate matter, and heavy metals, which can severely impact human and animal health. This process also produces carbon dioxide and other greenhouse gases, contributing to climate change. Besides, incineration generates toxic ash that must be disposed of in specialized hazardous waste landfills, posing further environmental and health risks. Moreover, incineration demands considerable energy, which can counterbalance some benefits gained from energy recovery [30]. Composting is suitable exclusively for organic waste and cannot process the extensive variety of inorganic waste generated by modern society. Ineffectively managed composting operations can produce methane, thereby contributing to greenhouse gas emissions. Additionally, composting can emit unpleasant odors and attract pests, which can be particularly problematic in urban areas [31]. In contrast to landfills, plasma pyrolysis does not generate leachate, thus eliminating the risk of soil and groundwater contamination. Plasma pyrolysis can reduce waste volume by up to 99 %, significantly diminishing landfill space needs. This technology can process various waste types [19,20], including hazardous, biomedical, electronic, and municipal solid waste, making it a versatile solution for contemporary waste management. By substantially reducing the volume of waste, plasma pyrolysis decreases the demand for landfill space and associated costs. In addition, the valuable by-products (e.g., syngas, metals) recovered from the process can be sold, providing additional revenue streams. The thorough breakdown of hazardous materials minimizes the risk of exposure to toxic substances for both workers and the surrounding community. The high temperatures in plasma pyrolysis ensure the complete destruction of pathogens, making it particularly suitable for treating biomedical waste. Unlike landfills and incineration, plasma pyrolysis does not require extensive long-term monitoring and management, thereby reducing long-term operational costs.

The dairy industry is mainly considered the primary source of wastewater generated from food processing, a significant problem some countries still face. In this regard, increasingly strict requirements for the waste treatment process have been imposed, considering the wastewater standards that have been developed in recent years. To examine such standards' efficiency, the industry's environmental footprints should be constantly traced and evaluated [32]. For instance, upon releasing wastewater into the environment, excessive oxygen consumption, eutrophication, impermeabilization, toxicity, and other repercussions recurrently threaten the surrounding discharge area [33]. Further, adding dairy waste such as milk to waste would lead to the loss of valuable products. Several methods have been established to dispose of dairy wastes, such as Physico-chemical and biological treatment methods, to name a few [34]. A study examined the production feasibility of biomethane from dairy manure. To this end, several appropriate technologies were employed to produce some eco-friendly energy sources and fuels from dairy wastes. In the designed sourcebook, several instructions were given to facilitate the exploration of the alternative usages of biogas produced from such waste type [35]. Another study [36] focused on different discharge sources of the dairy industry and the consequent environmental repercussions. They further suggested biological treatment methods for dairy waste fermentations. The handling of dairy waste poses substantial environmental and operational challenges for the dairy industry. Conventional treatment approaches, such as anaerobic digestion [37], composting [31], mechanical separation [38], aerobic treatment [39], and land application [29], have inherent limitations, including prolonged processing durations, odor generation, incomplete pathogen elimination, significant energy requirements, and extensive land use. Plasma pyrolysis, an advanced waste treatment technology, emerges as a promising alternative. Anaerobic digestion is a biological process that transforms organic waste into biogas and digestate. Despite its advantages, this method has several drawbacks [40]. A major concern is the production of methane, a potent greenhouse gas, during both controlled and uncontrolled phases, contributing to climate change. Furthermore, the process can be slow, necessitating long retention times to achieve substantial waste reduction.

The complexity and high cost of maintaining optimal conditions for microbial activity can also pose significant challenges, particularly in developing regions. Composting is a natural process that breaks down organic matter into nutrient-rich soil amendments. However, this method has its limitations [41]. It can emit odors and volatile organic compounds (VOCs), contributing to air pollution. Additionally, composting requires significant land and, if not properly managed, can attract pests. The final product may also contain pathogens, heavy metals, and other contaminants if the input waste is not adequately sorted and sanitized. Mechanical separation involves dividing waste into various components for further processing or recycling. While it effectively improves recycling rates, the method faces inefficiencies due to contamination and the heterogeneity of waste streams. This technology can be capital-intensive, and improper maintenance or operation may result in equipment failure. Moreover, the residual waste after

separation often still requires additional treatment, presenting an ongoing challenge [42]. Aerobic treatment [43], including processes like activated sludge, utilizes oxygen to break down organic pollutants in wastewater. The main disadvantages are high energy consumption due to the continuous need for aeration and the generation of excess sludge, which necessitates additional handling and disposal. Additionally, maintaining an optimal microbial environment requires precise control, rendering the system sensitive to variations in influent quality and environmental conditions. As previously mentioned, land application [29] involves dispersing organic waste or treated sludge on agricultural land as fertilizer. This method can result in soil and water pollution if the waste contains heavy metals, pathogens, or pharmaceutical residues. The long-term buildup of these contaminants can deteriorate soil health and pose risks to human health through the food chain. Besides, regulatory restrictions and public opposition limit the feasibility of land application in many regions.

For years, thermal plasma technology has been constantly developed [44]. In line with such development, several countries have set up laboratory and pilot-scale projects to efficiently utilize plasma pyrolysis as a potential technology for reproducing raw materials and some forms of energy from organic waste [45–50]. Plasma torches generated exceedingly high temperatures; hence, the best candidate to be applied to waste processing [51–53].

Presently, there is an existing gap in the plasma pyrolysis method for disposing of spoiled dairy, necessitating innovative approaches. This gap stems from insufficient comprehensive research and development to tackle the distinctive characteristics and challenges intrinsic to dairy waste management. Critical areas necessitating focus encompass optimizing plasma parameters for efficient dairy waste treatment, assessing the quality and safety of resulting by-products, and evaluating scalability and cost-effectiveness for broad implementation within the dairy sector. Addressing these gaps through targeted scientific investigation and technological advancement harbors substantial potential to augment dairy waste management protocols' sustainability and environmental robustness. The current research aims to delve into the disposal options of spoiled milk, which belongs to the category of wet waste. To date, no efficient solution has been provided to dispose of the waste under study but to release it into the environment. This research employed plasma pyrolysis to dispose of dairy waste efficiently, especially spoiled milk, and use the exhaust gases released from the waste gasification process to reproduce the fuel required for energy converters and turbines. Although dairy wastes are biodegradable, releasing them in nature, especially in running water (the simplest and most common method of getting rid of waste), will intensify the Chemical Oxygen Demand (COD) of water in the long run, hence consequent irreversible damages to the ecosystems [54–56]. In addition, the stink of such waste will make the surrounding disposal area unpleasant and sometimes uninhabitable. Applying the plasma pyrolysis system during disposal can significantly help overcome the mentioned problems. Dairy waste comprises organic constituents such as fats, proteins, and lactose. Plasma pyrolysis can thermally disintegrate these organic elements, transforming them into simpler gases and residues [20]. Plasma pyrolysis is recognized for its adaptability in addressing an array of waste categories, encompassing both hazardous and non-hazardous substances. This characteristic renders it well-suited for an extensive spectrum of waste streams, presenting a holistic resolution for varied waste disposal requirements. In contrast to conventional incineration techniques, plasma pyrolysis potentially provides reduced emissions benefits, given its operation within a controlled environment that minimizes the discharge of pollutants into the atmosphere [18].

Several methods have been established to apply thermal plasma to liquids. For example, in 1996, Schneider et al. examined how to dispose of acetone as a hazardous waste via a non-transferred plasma torch [57]. In addition, they explored the efficient liquid waste treatment options and studied the key parameters involved in plasma chemistry [58,59]. Westinghouse Plasma Corporation introduced a novel technology leading to the invention of a plasma pyrolysis reactor that could only perform the liquid waste conversion process, including polyphenol biphenyl, via a non-transferred plasma torch [60,61].

Several studies have been conducted in the literature to evaluate the possible disposal options. For instance, Capote et al. designed a device for liquid waste processing based on the waste characteristics that could be used to process bio-waste through gasification and waste sorting. They reported that this method successfully converted hazardous materials of different types into non-hazardous products that met most emission standards. The solid residue was either recycled or reused without producing fly ash, bottom ash, dioxin, or hazardous furan [62]. In another study, Nishioka et al. investigated the mechanism of organic compound decomposition using DC water plasmas with no external steam generator [63]. Karimi et al. [19] examined the effect of plasma pyrolysis on the spent catalyst and its conversion into value-added products via the transferred thermal plasma for the first time. According to their statements, plasma treatment proved efficient in producing hydrogen, carbon monoxide, methane, ethylene, etc., from the spent catalyst and preventing the formation of products with longer carbon chains of pernicious elements. In 2023, Aghayee et al. empirically demonstrated the viability of the transferred thermal plasma technology as a practical method for disposing of spent caustic. Their investigation, utilizing gas chromatography (GC) analysis, identified hydrogen (H_2) as the primary product, offering environmental benefits [20]. Kheyriyeh et al. demonstrated that plasma waste reactors can produce recycled energy as non-toxic renewable fuel, generating minimal environmental pollutants. Employing a thermal transferred arc plasma reactor, they conducted a feasibility assessment on the pyrolysis of three waste types: Antar, OTD, and Tar. Three distinct experiments were undertaken, investigating the pertinent parameters for the preparation and production of common gaseous raw materials, notably hydrogen, along with the quantification of their production yields [18]. Considering the background of plasma technology in waste management and the production of a significant amount of dairy waste, the idea of feasibility of disposal of this waste was proposed for the first time in this research. Therefore, with the design of a transfer thermal plasma reactor, the feasibility of eliminating this waste has been established. In this research, the data of several feasibility tests are presented. The production of very high temperature inside the electric arc and its interaction with the feed plays an important role in the optimization of these experiments. These tests show the challenges in scaling up this technology. Furthermore, the effectiveness of plasma pyrolysis in processing dairy waste was demonstrated through the substantial production of valuable syngas components like hydrogen and hydrocarbons, while minimizing harmful emissions. This aligns with the observed enhancement in syngas quality, making plasma pyrolysis a promising, eco-friendly waste management

solution. Thus, FTIR spectroscopy not only provided essential structural and quantitative data but also underscored the potential of plasma technology in sustainable waste processing, particularly in mitigating environmental impact in the northern regions of Iran.

1.1. Plasma pyrolysis

Plasma pyrolysis systems can be engineered with modular configurations, providing scalability to align with the waste disposal requirements across various scales, from small communities to larger urban areas [64]. These systems frequently integrate sophisticated control and monitoring technologies, guaranteeing meticulous control over process parameters and optimizing the efficacy of waste treatment. Plasma pyrolysis systems are probably the most novel systems that can eliminate some common defects in many cases. The approach used in this project highlights the application of plasma pyrolysis. Plasma pyrolysis furnaces are generally divided into two types: those with refractory walls (made of solid) and those with water walls. The refractory furnaces do not require air or additional moisture mainly due to utilizing several steam boilers whose required heat is provided by waste burning. The exhaust gasses must be cooled before being released into the ambient air; therefore, steam boilers facilitate the release of heat gasses produced through combustion. As a result, the ultimate temperature of the exhaust gasses will be reduced by half or even a third. In the second type of furnace, i.e., those with water walls, the internal wall of the combustion chamber comprises several vertically connected pipes, eliminating the need for additional air due to the steam recovery and temperature control inside the furnace. These furnaces are superior to those with refractory walls in terms of their design technology. However, the produced moisture inside these furnaces can be highly problematic mainly due to the possibility of the combination of the humidity with the gasses released from the plastic pyrolysis and other waste constituents and the formation of strong acids, hence corrosion and further severe damage to the pipes of the furnace walls. Waste incinerators are not only an effective solution to pollution reduction (if used properly based on the instructions), but they are cost-effective despite their relatively high construction costs [65]. In many disposal methods where a process such as incineration is carried out on the waste, the final waste or remaining ash will contain less environmental pollution. In some processes, such as plasma pyrolysis for waste treatment, which is also the main focus of this research, the final waste has gone one step further in productivity, facilitating their vast applications in various industries [66–68]. Owing to their structures, many of the slag (or slag-like) residues from the pyrolysis process are ideal for use in some industries, such as road and urban constructions, as well as glass and ceramic products [69–73]. These slag-like residues are of great economic significance mainly because, once sold, they will bring economic benefits to the plasma pyrolysis plants [74]. In addition, maximum and efficient reuse of waste material and elemental recycling are the other benefits that significantly contribute to environmental safety.

A feasibility study is defined as applying plasma pyrolysis, considering the relatively accurate input waste quantity and quality information, and predicting the alterations during the designed period, i.e., the gasifier service life. Among the significant factors determining the type and capacity of the plasma pyrolysis unit and its production are the amounts, composition, and calorific value of the treated waste that should be specified through a detailed analysis of their current conditions and prediction of their future.

The two most significant plasma generation methods used for waste processing are direct-current plasma torches (transferred and non-transferred) and Radio-Frequency (RF) Inductively Coupled Plasmas (ICPs) [51,75,76]. Two main characteristics have made the application of plasma torches, produced by either transferred or non-transferred methods, efficient in the plasma pyrolysis process: their lack of involvement and excessive changes throughout the mentioned processes. Transferred arc refers to a state where an arc is transferred from the electrode to the workpiece [77,78]. On the contrary, a non-transferred arc is when the plasma arc is internally formed, and the plasma jet comes out of the torch. For years, thermal plasma technology has been widely used to dispose of various hazardous and toxic wastes [79–82]. Before the application of this technology to incinerators, considerably excessive amounts of dioxins, furans, and sometimes heavy metal residues left as the incinerator outputs have caused a serious problem, thus making such systems life-threatening [83,84].

Moreover, a quenching system after the chamber led to a quick temperature decrease and significantly reduced dioxin and furan release. Other exhaust gasses can dramatically be reused for energy generation purposes. For instance, hydrogen, methane, and other hydrocarbons can be used as supplementary fuels for power plants. The waste must be treated through preliminary sorting before the combustion process in several previous generations of incinerators. However, the plasma system no longer feels the need for a primary sorting and recycling process before combustion. In addition, the high plasma temperature facilitates waste disposal, regardless of their types and constituents; hence, there are no further concerns. Compared to its counterparts, the plasma pyrolysis system for waste treatment has been acknowledged as the most environmentally friendly method due to its utilization of plasma torches and pyrolysis. It enjoys other advantages such as less landfill need and consequently fewer leftover transportation and land costs during the waste burial stage [83,84]. Though, it is crucial to acknowledge that the merits and demerits of plasma pyrolysis can exhibit variability contingent upon distinct waste compositions, technological enhancements, and regulatory frameworks. The ensuing delineation enumerates certain drawbacks associated with this approach. Plasma pyrolysis necessitates a substantial energy input to maintain the elevated temperature of the plasma. This energy requirement may be sourced from non-renewable reservoirs, potentially compromising environmental advantages. Nevertheless, utilizing renewable energy resources to drive plasma generation can alleviate this drawback. The primary capital investment for the establishment of a plasma pyrolysis facility may be relatively elevated when juxtaposed with certain conventional waste management approaches. The efficacy of plasma pyrolysis is contingent upon the specific composition of the waste, with some waste types presenting challenges for optimal processing. Notwithstanding its ongoing development, plasma pyrolysis might encounter constraints in widespread commercialization compared to more entrenched waste management methodologies.

2. Materials and methods

Milk is a substance with a liquid structure that shares significant similarities with the molecular structure of water. Approximately 80 % of milk is composed of H_2O (water). In contrast, the rest is characterized by milk's hydrocarbon structures, fats, and acids. In addition, some types of glucose and proteins are also found in the milk structure. Most foods undergo some physical and chemical changes over time that are usually regarded as spoilage, and milk and other dairy products are no exception; hence, they are susceptible to spoilage.

Fats in milk are the most important parameter in determining the nutritional value of milk. Chemical structures all belong to the category of hydrocarbons. The sugar in milk is mainly composed of lactose, and during the spoilage process of milk, it will turn into lactic acid, and its molecular bonds will change. Normally, milk turns sour and spoils very quickly. The reason for this change is the structural transformation of citric acid in milk into acetic acid and carbon dioxide. In general, during the process of souring and spoilage, the molecular structure of milk does not change much. The chemical composition of the waste tested in this research was not much different from normal milk, so the range of molecules in spoiled milk can be considered similar to regular milk. However, a

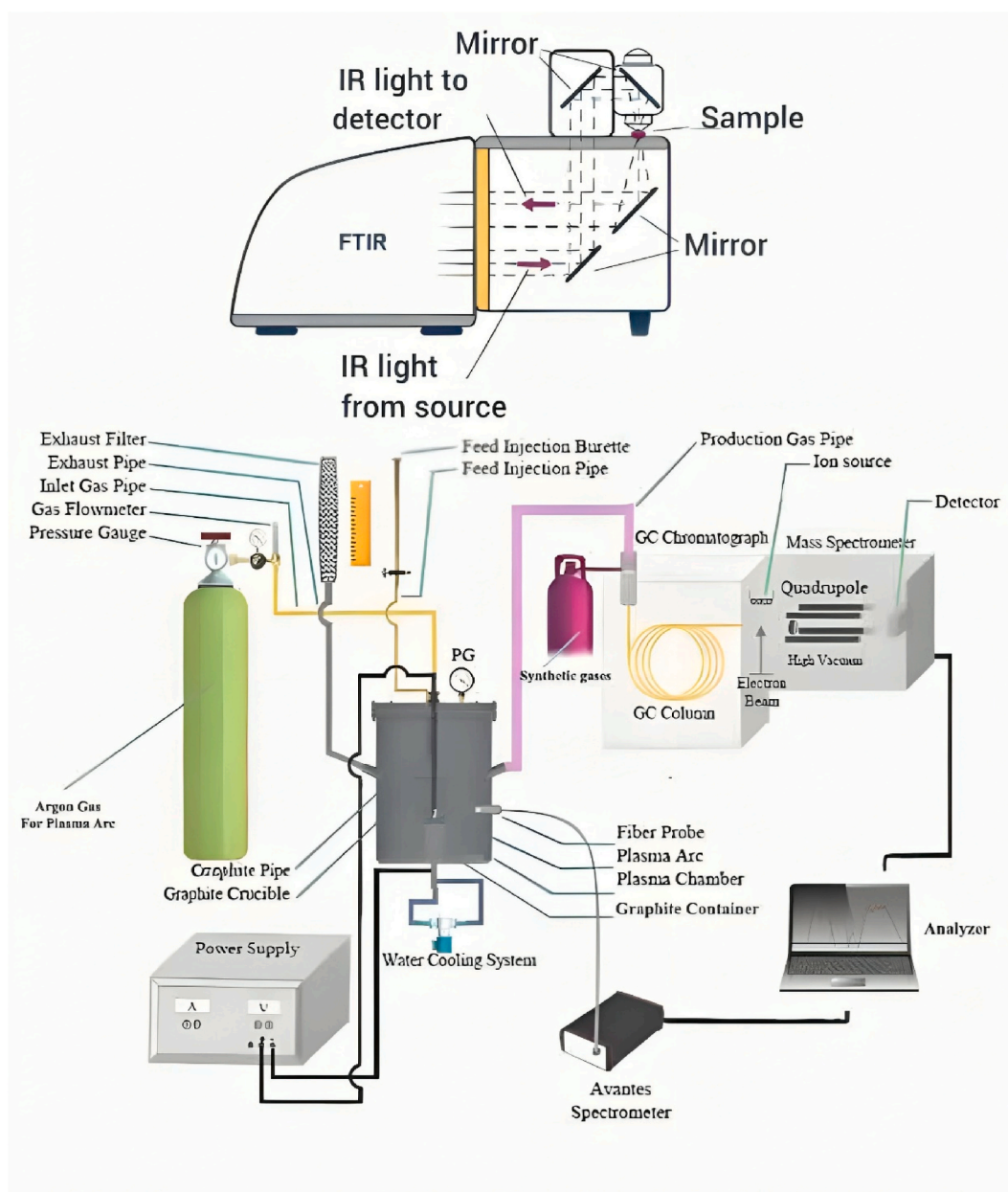


Fig. 1. The laboratory setup implemented in this study.

sample of spoiled and regular milk was examined to further ensure this information. FTIR analysis was used to compare these two samples. An infrared spectrophotometer, SpectrumOne, equipped with a ZnSe detector and manufactured by PerkinElmer (USA), was utilized. In this research, tests such as CNH (Carbon-Nitrogen-Hydrogen) analysis and SEM (Scanning Electron Microscopy) appeared to be effective. Nevertheless, conducting these tests was considered unnecessary due to the anticipated outcomes. It is advisable to perform an analysis such as mass spectrometry with an electron microscope to determine the mass and the specific compounds of each element in the material. However, for the substance in this project, given that milk is a liquid, it was necessary to first create a solid tablet of the milk by drying it prior to conducting this test. This procedure would lead to the loss of a significant portion of the hydrogen, as it is present in the composition of water. Since hydrogen was a key parameter in the experiments, its absence would compromise the validity of the data.

The current research employed argon as the constituent plasma gas. Its low ionization energy has made it more feasible; hence, it is a good candidate for obtaining a plasma of acceptable power with less energy consumption. The constant gas flow in all experiments was considered to be 10 l/min. Throughout this study, a total number of four experiments were carried out. In the first and second experiments with the voltage and current of 35V and 170A, respectively, the device's power was measured as 5950W. In the third and fourth experiments with the voltage and current of 30V and 150A, the power was obtained as 4500W. During these experiments, the device first operated without feed injection for 1 min with an empty arc so that the temperature of the furnace and electrode would increase to a specific level. The thermal arc (Arc discharge with a current of more than 30 A according to Ref. [85]) plasma reactor comprises a cylindrical chamber made of stainless steel, graphite-carbon electrodes (anode), graphite crucible (cathode), water-cooling system for the cathodic region, quartz tube, stainless steel chamber door, steel fasteners, feedstock inlet pipe, and gas outlet pipes. The anode is a long cylindrical tube with a length of 45 cm, an internal radius of 3 mm, and an external radius of 6 mm. The anode's inner channel introduces the input gas into system. A substantial portion of this electrode is made of pure carbon (over 80 %). The rationale for choosing a graphite-carbon material for the anode is that it possesses high electrical conductivity, heats up more slowly, and, due to its robust structure, is more resistant to corrosion. Nevertheless, the electrode tip was eroded in each experiment. Consequently, the gap between the cathode and anode was readjusted after each experiment. Corrosion of carbon in the graphite electrode can occur due to heat and also in the presence of oxygen in the feed. As a result, it is possible to produce monoxide and carbon dioxide gases along with other products, which is inevitable. Owing to the steel's thermal tolerance, which extends to around 1400 °C, and considering the relatively small volume of the plasma flame, there is not a notable increase in temperature within the chamber walls. It is important to note that the experimental duration, spanning from the plasma ignition to its cessation, is limited to a maximum of 4 min. During this period, the overall heat transfer from the gases inside the chamber to the walls remains relatively modest. Consequently, it can be deduced that the chamber does not incur thermal damage.

To enhance the conductivity of the crucible into which the solution was injected, we put some metal shavings (dishwashing wire) in it to fasten the formation of the plasma arc and facilitate its direct application to the container containing the spoiled milk. Of note, the electrode gap up to the crucible surface was considered 1 cm, and at the same time, argon gas was injected into the chamber with a constant flux of 10 l/min to help form plasma. All four experiments were carried out under the same conditions, processes, and instructions. Fig. 1 shows the laboratory setup implemented in this study. The feedstock inlet conduit is a 20-cm-long cylindrical plastic tube with an internal radius of approximately 3 mm. The selection of this tube is based on the convenience of its availability and, notably, its thermal resilience. The utilization of a tube with this radius is warranted due to the liquid state of the feedstock, ensuring accurate and uniform feedstock injection into the thermal arc plasma center. This ensures a continuous and uniform injection of feedstock. Tubes featuring larger radii resulted in the injection of droplets and intermittent flow, thereby disrupting the homogeneity of experimental conditions. Nevertheless, beyond the reactor, a 20-cm-long rubber hose with an internal diameter of 5 mm was utilized for conveying the experimental liquid. This was done to prevent the clogging of the feeding system caused by crud in the solution, which occasionally persisted despite efforts to smooth them out within the test tube. In the first experiment, the spoiled milk was injected into the furnace with a relatively uniform flow. Since milk curdling is highly probable, which may increase the risk of the injection tube getting clogged, shake the solution before adding it to loosen the curds as much as possible. The consumed feed was about 300 ml. In order to uphold atmospheric pressure, numerous outlets have been incorporated into the chamber's walls and door for the release of internal gases. These outlets, situated external to the chamber, capitalize on pliable silicone hoses to guide the chamber gases outside the experimental setup, given the ambient temperature remains relatively low. Furthermore, one of these outlets collected samples in balloons that had been pre-evacuated of air. Once the Plasma pyrolysis of the waste process was completed, some of the exhaust gas was collected in a balloon and delivered to the gas chromatography laboratory to evaluate its constituent elements. The fundamental principle of GC entails vaporizing the sample and introducing it into a chromatograph, traversing a stationary phase. During its progression through the stationary phase, distinct components undergo differential interactions, resulting in their separation [86–88]. The gaseous products were injected from the balloon into the gas chromatograph (GC; Agilent-7890A, column HP-Al/S) with an FID detector for the detection of CH₄, C₂H₄, C₂H₆, C₃H₆ and so on, thermal conductivity detector TCD-1 for the detection of N₂, Ar, CO, and CO₂, and the detector TCD-2 for the detection of H₂ and He.

As we mentioned before, the elevated temperature conditions facilitate the thorough decomposition of organic substances, encompassing intricate compounds in spoiled milk. The procedure effectively disintegrates organic matter into simpler gaseous components, leaving no residual substances. Moreover, in contrast to conventional disposal approaches, plasma pyrolysis presents a potentially more ecologically benign alternative. The elevated temperature procedure can alleviate the emission of detrimental pollutants and reduce the environmental repercussions of spoiled milk disposal. Besides, the elevated temperatures attained during plasma pyrolysis play a role in eradicating pathogens and microorganisms found in spoiled milk, thereby establishing a hygienic and secure disposal technique. This holds particular significance within the realm of food waste management. Furthermore, plasma pyrolysis exhibits the potential to markedly diminish waste volume, particularly in the case of spoiled milk characterized by elevated

water content. Through processing, the residual materials can be minimized, enhancing the efficiency of waste management processes. Plasma pyrolysis technology is scalable, enabling effective disposal solutions applicable to small-scale and large-scale operations. This adaptability allows for efficiently managing diverse quantities of spoiled milk in various settings.

3. Results and discussion

FTIR spectroscopy delivers comprehensive insights into the functional groups within a sample, facilitating the identification of specific molecules. The resulting spectra offer valuable structural information, including details on bond types, functional groups, and molecular conformations. Additionally, FTIR enables quantitative analysis by measuring the absorbance of infrared radiation at various wavelengths. This capability is invaluable in environmental science, food analysis, and quality control processes in industries like manufacturing and pharmaceuticals. FTIR analysis is non-destructive, enabling the examination of samples without altering or damaging them. It yields rapid results, supporting high-throughput analysis of large sample sets in a short time. This spectroscopy is versatile and applicable to a wide array of sample types, including solids, liquids, gases, thin films, and various states such as powders, solutions, and gels. FTIR spectroscopy is relatively cost-effective compared to other methods like nuclear magnetic resonance (NMR) spectroscopy or mass spectrometry. The results obtained from the FTIR test in the spoiled milk waste show that all the extremum points of the wavelength have increased by almost 100 units (Fig. 2(a) and (b)). This coordination in moving to the left indicates that almost all the components have changed slightly, and only at some points, covalent changes took place for hydrogen-carbon bonds. In both graphs, the longest wavelength is related to bonds of the O-H, showing Water as the main component in milk. More precisely, in the sample of regular milk, where this peak occurred at a lower wavelength, there were weaker carboxylic acid compounds, which was expected. The other two peaks between 2800 and 3000 cm^{-1} include the C-H hydrocarbon bond. The next peak in both graphs occurred at almost the same point with the wavelength of 2078 cm^{-1} , representing the weak $\text{C} \equiv \text{C}$ triple bond. The deepest peak of the two graphs in the figure is related to the $\text{C}=\text{C}$ bond structure, and the next peak of this bond has become a strong six-membered carbon ring

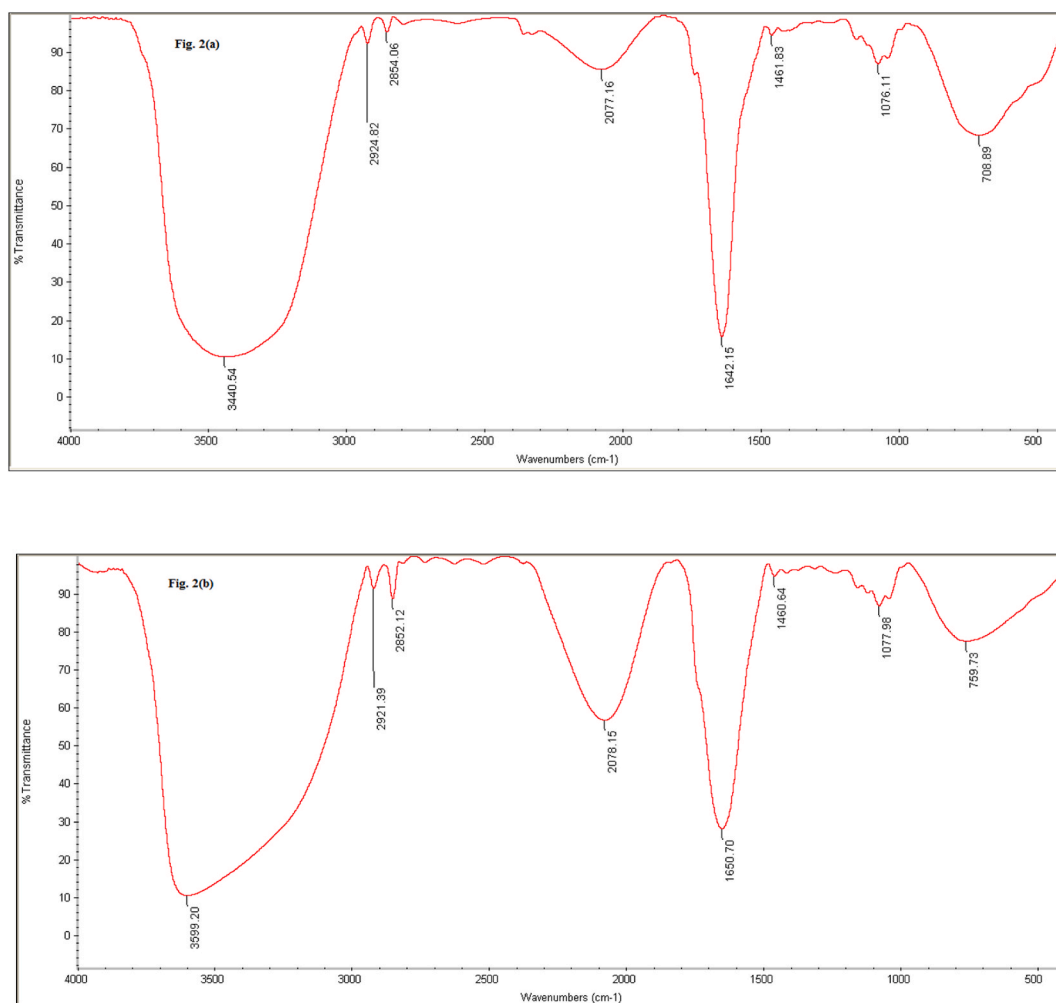


Fig. 2. (a) and (b) ATR-FTIR spectra for regular and spoiled milk, respectively.

bond. The peak corresponding to the wavelength of 1460 is also related to carbon-hydrogen rings. This time, the bonds are shared and of weak type and are classified in aromatic structures. The last two peaks, which have the lowest wavelengths, include carbon ternary structures and mono-carbon halides, respectively [89]. In general, the changes made during the corruption process are not of a type that will cause much change in the results obtained from the plasma waste incinerator.

As observed in Table 1, the results from the first experiment indicated that disposal of dairy waste using plasma pyrolysis is possible. Still, the final products and obtained synthetic gases are very desirable. In addition, no trace of hazardous substances was detected in the results. Instead, some useful products were obtained. For instance, hydrogen and hydrocarbons can be used in power plant fuel systems. Carbon monoxide and carbon dioxide are among the gases with less efficiency. Of note, this method's ratio of hydrogen to carbon monoxide (syngas), an important parameter that determines the syngas quality, was much higher than in other methods, thus making the plasma system the best option for waste disposal with the least pollution. Syngas finds application in heating processes, where the heat generated from combustion is harnessed for industrial operations or space heating. It functions as a precursor in the production of diverse chemicals, such as methanol and ammonia, necessitating customization of its composition to suit the distinctive requirements of each chemical synthesis process. Additionally, it serves as a viable fuel for specific vehicle types, particularly those featuring engines designed for gaseous fuels. Careful consideration of composition parameters is imperative to ensure optimal combustion efficiency in this application. Elevated syngas production proves advantageous in instances of substantial demand, particularly in applications such as industrial processes, power generation, or hydrogen production. The utilization of syngas as a cleaner substitute for conventional fuels holds potential for mitigating environmental impact, particularly when displacing energy sources characterized by higher carbon intensity [90].

Table 1
The measured data for the production of products.

Number of experiments	1	2	3	4
Voltage (V)	35	35	30	30
Current (A)	170	170	150	150
Power (Kw)	5.95	5.95	4.5	4.5
Feed (L)	0.3	0.3	0.3	0.3
H ₂				
%	60.396	58.699	57.919	57.407
SLM	3.169	0.442	2.260	2.480
CO				
%	27.883	28.287	30.779	30.486
SLM	1.463	0.213	1.201	1.317
CO ₂				
%	10.292	11.554	7.458	8.426
SLM	0.540	0.087	0.291	0.364
Methane				
%	0.896	0.664	2.409	2.060
SLM	0.047	0.005	0.094	0.089
Ethane				
%	0.019	0	0.051	0.093
SLM	0.001	0	0.002	0.004
Ethylene				
%	0.286	0.266	0.974	0.972
SLM	0.015	0.002	0.038	0.042
Propane				
%	0.019	0	0.026	0.046
SLM	0.001	0	0.001	0.002
Propylene				
%	0.019	0.040	0.077	0.116
SLM	0.001	0.0003	0.003	0.005
n-Butane				
%	0.191	0.398	0	0
SLM	0.010	0.003	0	0
n-Pentane				
%	0	0.040	0	0.023
SLM	0	0.0003	0	0.001
Iso-butane				
%	0	0	0.308	0.347
SLM	0	0	0.012	0.015
Iso-butene				
%	0	0	0	0.007
SLM	0	0	0	0.0003
Iso-Pentane				
%	0	0	0	0.009
SLM	0	0	0	0.0004
Total hydrocarbons				
%	1.429	1.461	3.844	3.657
SLM	0.075	0.011	0.150	0.158

During waste interaction with plasma, the waste is introduced into the plasma environment, experiencing direct exposure to high-temperature active species. Consequently, diverse chemical reactions occur. The characteristics of the waste, including its chemical composition and the plasma parameters, such as the quantity and type of active species, as well as temperature, are paramount in determining the specific nature of these reactions [19]. In the first two experiments with the voltage and current of 35V and 170A, the amounts of produced hydrogen were 60.396 % and 58.699 %, respectively. The same values in the third and fourth tests, when the voltage was reduced by 30V and the current by 150A, were equal to 57.919 % (2.260 SLM) and 57.407 % (2.480 SLM), respectively. The amounts of carbon dioxide (in percentage) in all four experiments were obtained as 10.292 % (0.540 SLM), 11.554 % (0.087 SLM), 7.458 % (0.291 SLM), and 8.426 % (0.364 SLM), respectively. According to Table 1, fluctuations in the voltage and current would change the production percentage of carbon monoxide and hydrocarbons. To be specific, the production percentage of carbon monoxide changed from 27.883 % (1.463 SLM) to 30.779 % (1.201 SLM), and that of hydrocarbons from 1.429 % (0.075 SLM) to 3.844 % (0.150 SLM).

To better analyze the data provided in Table 1, Fig. 3(a) and (b) list the percentages of the amounts of products in different samples. The effect of variations in the voltage and current on the production amount of hydrocarbon products in each experiment was evaluated based on the results, and it can be concluded that the effect of each variation was selective, thus yielding different results for each product. The variation rates for methane ranged from 0.664 % (0.005 SLM) to 2.409 % (0.094 SLM), ethane from 0 to 0.093 % (0.004 SLM), ethylene from 0.266 % (0.002 SLM) to 0.974 % (0.038 SLM), propane from 0 to 0.046 % (0.002 SLM), propylene from 0.019 % (0.001 SLM) to 0.116 % (0.005 SLM), isobutane from 0 to 0.347 % (0.015 LM), and n-butane from 0 to 0.398 % (0.003 SLM). The percentage of n-pentane production varied from 0 to 0.040 % (0.0003SLM). However, the production percentages of isobutene and isopentane were equal to zero in all experiments except for the fourth one at 35V. Their SLM values were quite small. The first experiment with a voltage of 30V and a current of 150A exhibited the best plasma effect in yielding the maximum percentage of hydrogen production. Fig. 4 shows the different product rates obtained from different experiments in SLM.

The feed's plasma processing byproducts, including hydrogen, carbon dioxide, carbon monoxide, and hydrocarbons, are shown in Fig. 5. Flame Ionization Detector (FID) detects hydrocarbons, while Thermal Conductivity Detector detects carbon dioxide, carbon monoxide, and hydrogen (TCD). Based on the results of experiment 1, which were 20.785 (mol%) H₂, 9.594 (mol%) CO, 3.539 (mol%) CO₂, 0.311 (mol%) Methane, 0.006 (mol%) Ethane, 0.098 (mol%) Ethylene, 0.004 (mol%) Propane, 0.009 (mol%) Propylene, and 0.063 (mol%) n-Butane values, the mole value in this figure was calculated.

The device conditions were changed in the third and fourth experiments, which were performed with a lower device power, i.e., 4500W. Different factors can determine whether or not the process that is about to yield some gases is appropriate. One of the determining parameters in the processes that lead to the production of some gases, such as hydrogen and greenhouse gases, is the ratio of the produced gases to their counterparts. In this respect, some processes, such as waste incineration and other combustion processes, are usually criticized for their excessive production of harmful gases compared to the harmless ones released into the ecosystem.

In most incineration processes and even the traditional disposal methods such as landfills, the gases released into the atmosphere contain a relatively high volume of harmful and greenhouse gases. Take methane, for example. It is released into the atmosphere without any processing in the landfill processes. Despite the importance of methane as a valuable fuel, its uncontrolled emission and consequent release into the atmosphere will have severe environmental repercussions, such as air pollution and global warming [91]. Sometimes in other processes, some gases such as NO_x [92] and CO, which are often formed due to incomplete combustion, are abundantly produced, thus making the process inappropriate due to their role in intensifying environmental pollution. As already suggested, a suitable method to ensure the efficiency and reliability of the plasma pyrolysis process is to compare the amount of hydrogen produced in the process with that of carbon monoxide. In most combustion processes, the corresponding value for this comparison is less than one; hence, the pollution load will be higher. CO₂ and CO are separate yet interconnected variables frequently deliberated within the domains of environmental surveillance, combustion procedures, plasma pyrolysis, and air quality evaluations. Within the framework of plasma pyrolysis, the CO₂ and CO parameters represent pivotal factors, offering perspectives on process efficiency, the character of ongoing chemical reactions, and potential environmental repercussions. CO emerges as a prevalent

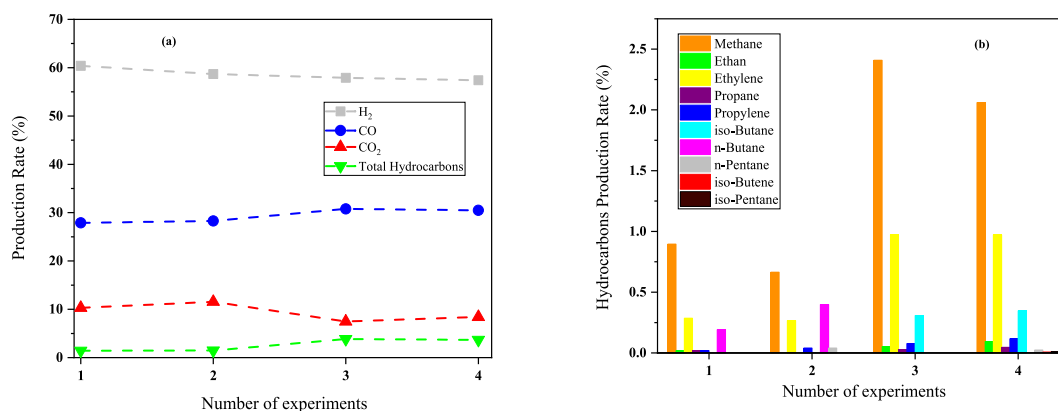


Fig. 3. (a) and 3(b) The production rate percentages of the four experiments.

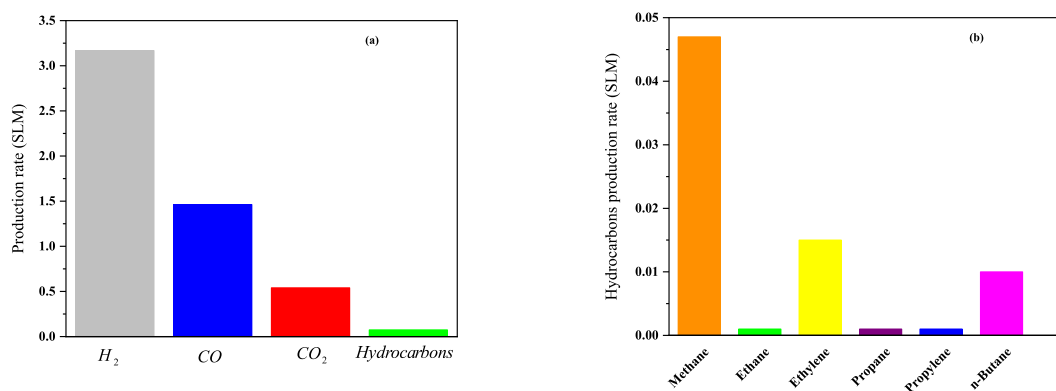


Fig. 4. (a) and (b) Production rate at $I = 170\text{A}$, $V = 35\text{V}$, and 0.3 L feed injection for sample 1.

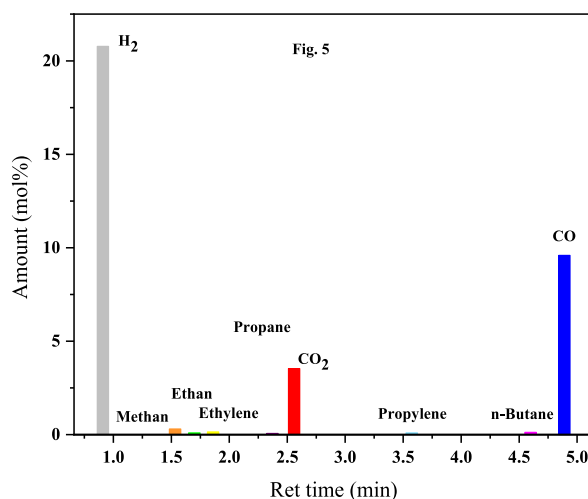


Fig. 5. The gas chromatography of output products from the 1st sample with FID and TCD.

intermediary product in the course of plasma pyrolysis. CO₂ and CO concentrations are quantified in plasma pyrolysis procedures through diverse analytical methodologies, such as infrared spectroscopy, gas chromatography, or dedicated sensors engineered for environments with elevated temperatures. As we mentioned before, in plasma pyrolysis, the gasification rate ($\text{H}_2 + \text{CO}$) denotes the effectiveness with which carbonaceous substances convert into syngas. This mixture commonly encompasses hydrogen and carbon monoxide alongside other gases. Plasma pyrolysis, a thermochemical procedure, employs elevated-temperature plasma to break down organic substances into elemental constituents or more straightforward compounds. The gasification rate in plasma pyrolysis, particularly concerning the production of H₂ and CO, is subject to the influence of various factors. The characteristics of the plasma, encompassing temperature, power, and residence time, significantly influence the gasification rate. Elevated temperatures and

Table 2

The syngas rate (H_2/CO), CO_2/CO , gasification rate ($\text{H}_2 + \text{CO}$), and combustion rate ($\text{CH}_4 + \text{CO}_2$) for experiments.

Number of experiments	1	2	3	4
Voltage (V)	35	35	30	30
Current (A)	170	170	150	150
Power (Kw)	5.95	5.95	4.50	4.50
Feed (L)	0.3	0.3	0.3	0.3
H ₂ /CO	2.166	2.075	1.882	1.883
CO ₂ /CO	0.370	0.409	0.242	0.276
H ₂ + CO				
%	88.279	86.986	88.698	87.893
SLM	4.632	0.655	3.461	3.797
Methane + CO ₂				
%	11.188	12.218	9.867	10.486
SLM	0.587	0.092	0.385	0.453

prolonged residence times typically promote heightened gasification. The makeup of the plasma gas, encompassing the variety and concentrations of reactive species, impacts the reaction pathways and, consequently, the gasification rate. Distinct plasma pyrolysis technologies, such as atmospheric pressure, microwave, or arc plasma, may demonstrate discrepancies in efficacy and gasification rates. As per the outcomes delineated in Table 2, the syngas yield in experiments 1 and 2, where the voltage and current were held constant at 35V and 170A, respectively, amounted to 2.166 and 2.075 correspondingly. In the third and fourth experiments, a reduction in voltage and current (to 30V and 150A) was implemented. This reduction exerted an influence on the generated syngas, resulting in a decrease of the targeted parameter to 1.882 and 1.883, respectively. The ratio of CO_2/CO was also scrutinized in this table, yielding values of 0.370, 0.409, 0.242, and 0.276 for the four experiments, respectively. The $\text{H}_2 + \text{CO}$ parameter was assessed in percentage and SLM (Standard Liters per Minute). Under a voltage of 35V and a current of 170A, the percentage of this parameter in the initial and subsequent experiments was 88.279 % and 86.986 %, respectively. Correspondingly, these figures were computed as 4.632 and 0.655 SLM, respectively. In the third and fourth experimental trials, the observed percentages exhibited minimal alterations and were, sequentially, 88.698 % (equivalent to 3.461 SLM) and 87.893 % (3.79 SLM). Utilizing methane and carbon dioxide combustion within the context of plasma pyrolysis finds application in waste management, the synthesis of syngas, and the generation of thermal energy or electrical power. The concluding parameter examined in Table 2 pertains to the quantity of $\text{CH}_4 + \text{CO}_2$. The minimal value for this parameter, expressed as a percentage, is 9.867 %, derived from the outcomes of the third experiment, whereas the maximal value is 12.218 %, recorded in the second experiment. Fig. 6(a) and (b) confirmed the effect of variations in the current and voltage on the syngas production. Specifically, as we mentioned before, upon increasing the voltage from 30 to 35V and current from 150 to 180A, the order of syngas would also increase. In this current project, the acquired data demonstrate notable consistency with the findings presented in our previous works [18–20].

To gain a more profound understanding of the product production process, the results were analyzed and reported based on energy density (Fig. 7(a) and (b)). This precise analytical method allowed for improved observation and comprehension of the changes and their impacts. The studies conducted revealed that increasing energy density leads to a rise in hydrogen production percentage and an increase in the syngas production rate. These findings suggest optimizing energy density can significantly enhance production processes and efficiency.

The calorific value is effectively categorized into two primary divisions: Higher Heating Value (HHV) and Lower Heating Value (LHV). The Lower Heating Value (LHV) is a crucial metric for evaluating the energy potential of syngas produced through plasma pyrolysis. Tables 3 and 4 examine the LHV and HHV of syngas generated from the plasma pyrolysis of dairy waste, utilizing gas chromatography to determine its composition. The LHV is calculated based on the volumetric concentrations of the component gases and their respective energy contributions. The results highlight the importance of LHV in optimizing energy recovery and improving the efficiency of plasma pyrolysis systems. By comprehending the LHV, more effective strategies can be devised for waste management and sustainable energy production, demonstrating the dual advantages of this technology in both environmental and energy sectors. Higher Heating Value (HHV) denotes the aggregate heat liberated during the thorough thermal breakdown of the feedstock substance within the plasma reactor, occurring under stable pressure. In summary, in plasma pyrolysis, both HHV and LHV offer perspectives on the energy content of the feedstock material as it undergoes thermal decomposition within the plasma reactor. HHV encompasses all released energy, encompassing subsequent reactions and condensation, whereas LHV offers a more cautious estimation, concentrating solely on the primary decomposition process [93].

In the realm of plasma pyrolysis, the concept of “cold gas efficiency” pertains to how efficiently inert gases such as nitrogen or argon are utilized to decompose waste materials into valuable outputs. Cold gas is injected into the plasma reactor chamber to establish conditions favorable for high-temperature reactions, simultaneously inhibiting undesired chemical reactions with the waste feedstock. The effective utilization of cold gas is imperative for sustaining stable plasma conditions and fostering desirable pyrolysis reactions, such as converting organic compounds into syngas or synthesis gas. Variables affecting cold gas efficiency encompass gas flow rate, gas purity, and the reactor configuration of gas injection systems. Enhancing cold gas efficiency in plasma pyrolysis procedures improves system performance by optimizing the conversion of waste materials into valuable products, concurrently reducing energy

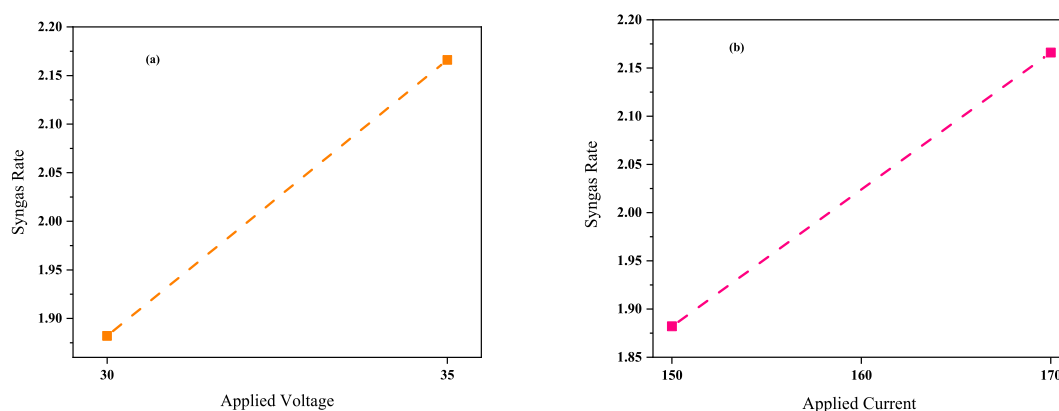


Fig. 6. (a) Syngas rate versus voltage and 6(b) Syngas rate versus current.

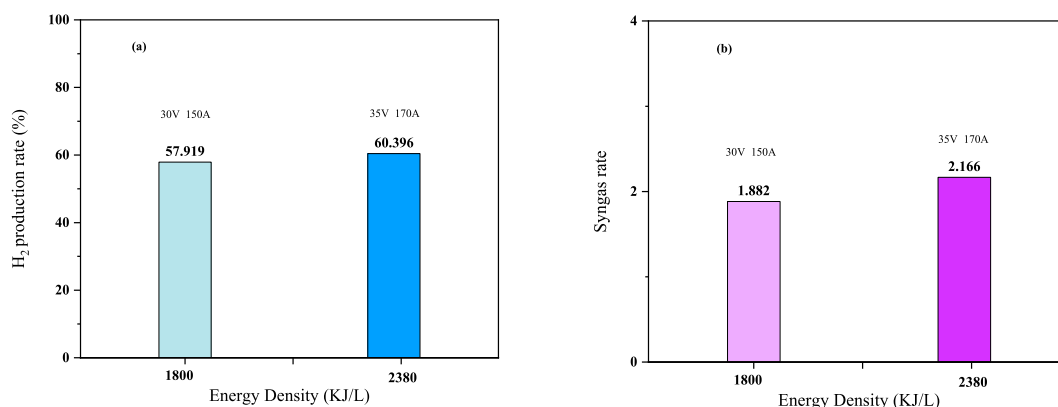


Fig. 7. (a) Hydrogen production rate and 6(b) Syngas rate versus Energy density.

Table 3

The LHV value of experiments.

LHV of experiments (MJ/m ³)	1	2	3	4
H ₂	6.523	6.340	6.255	6.200
CO	2.816	2.857	3.109	3.079
CO ₂	0	0	0	0
Methane	0.321	0.238	0.862	0.738
Ethane	0.012	0	0.033	0.059
Ethylene	0.170	0.158	0.578	0.576
Propane	0.018	0	0.024	0.043
Propylene	0.014	0.030	0.057	0.085
n-Butane	0.213	0.445	0	0
n-Pentane	0	0.043	0	0.025
Iso-butane	0	0	0.323	0.363
Iso-butene	0	0	0	0.007
Iso-Pentane	0	0	0	0.010

Table 4

The HHV value of experiments.

HHV of experiments (MJ/m ³)	1	2	3	4
H ₂	7.670	7.454	7.356	7.291
CO	3.513	3.564	3.878	3.841
CO ₂	0.916	1.028	0.664	0.750
Methane	0.357	0.264	0.959	0.820
Ethane	0.014	0	0.036	0.066
Ethylene	0.186	0.173	0.633	0.632
Propane	0.019	0	0.026	0.047
Propylene	0.015	0.032	0.062	0.094
n-Butane	0.235	0.490	0	0
n-Pentane	0	0.049	0	0.028
Iso-butane	0	0	0.364	0.410
Iso-butene	0	0	0	0.007
Iso-Pentane	0	0	0	0.011

consumption and operational expenses. This efficiency is crucial for extensively adopting plasma pyrolysis as a sustainable approach for waste management and resource retrieval [94]. This paper's maximum cold gas efficiency reaches 52.47 %, corresponding to the first experiment.

In the northern areas of Iran, where domestic, medical, and industrial waste is deposited within the central regions of the forests encompassing the provinces of Gilan, Mazandaran, and Golestan, there is a discernible risk of introducing substantial volumes of toxic leachate into the water and soil of these three provinces. This situation poses a significant crisis for the northern provinces of the country. Prospective initiatives might involve the integration of plasma pyrolysis technology in the aforementioned provinces. Last but not least, we utilized plasma pyrolysis to treat dairy waste, characterized by its predominantly liquid nature. Upon subjecting the dairy waste to the plasma pyrolysis process, we observed a notable outcome: the absence of any discernible residue. This outcome underscores the efficacy of plasma pyrolysis in decomposing organic materials, as the process effectively disintegrated the dairy waste,

leaving behind no residual components.

4. Conclusion

Plasma pyrolysis provides numerous advantages compared to conventional disposal approaches for dairy waste, fostering sustainable waste management practices. Its versatility enables the effective treatment of diverse waste types, including dairy waste with varying compositions, establishing it as a viable solution for sustainably managing a range of waste streams. Operating as a closed-loop system, this technology allows for utilizing produced gases and residues within the system or for other beneficial applications. This closed-loop characteristic augments the overall efficacy of waste treatment and enhances resource utilization efficiency. This study suggests plasma pyrolysis as a novel method to dispose of dairy waste and its conversion into value-added products. The spoiled milk in this research is a case study to assess the feasibility of this process. The results have shown that plasma pyrolysis can dispose of this hazardous waste and be considered a novel approach to the green synthesis of hydrogen from dairy waste. Experimentally decreasing the power of the plasma pyrolysis system by a decrease in voltage and current shows a decrease in hydrogen production and syngas rate. So, higher power causes high temperatures produced by plasma pyrolysis, leading to the strong breaking of molecular bonds in dairy waste. As an indicator of syngas quality, the syngas rate or H_2/CO ratio of 2.166 has been achieved, a desirable ratio among other methods used for syngas production. The H_2 production of 60.4 % is a noticeable amount, which is a promising result for a laboratory-sized project to examine its feasibility.

Statements and declarations

We confirm that the manuscript has been read and approved by all named authors and that there are no other persons who satisfied the criteria for authorship but are not listed. We further confirm that the order of authors listed in the manuscript has been approved by all of us.

Competing interests

All authors certify that they have no affiliations with or involvement in any organization or entity with any financial interest or non-financial interest in the subject matter or materials discussed in this manuscript.

Data availability

All data associated with this study is available on request.

CRediT authorship contribution statement

M. Fasihi: Investigation, Formal analysis. **B. Mohammadhosseini:** Validation, Project administration, Conceptualization. **F. Ostovarpour:** Writing – original draft. **M. Shafiei:** Validation, Formal analysis. **M.S. Abbassi Shanbehbazari:** Writing – review & editing. **M. Khani:** Writing – review & editing, Methodology. **B. Shokri:** Resources.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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